

Discussion 1: AI and Employment AI and Social Adaptation

Discussion Topic:

Businesses, governments, and academic institutions around the globe are heavily investing in AI technologies. These tools are revolutionizing almost every industry and academic discipline. As a result, the nature of work, governance, and scientific discovery will change in unprecedented ways. AI technologies will reshape every aspect of the economy in the next decade. We have the capability of completely automating the supply chain.

- How do you envision the integration of AI technologies into businesses impacting the nature of jobs and the composition of the workforce?
- How might societies need to adapt to the increasing integration of AI technologies into everyday life?

My Post:

Hello Class,

Society is at the dawn of a new technological revolution fueled by AI, an AI revolution. AI is such a powerful transformational technology that its integration into society has the potential to fundamentally change how individuals work, communicate, make decisions, and conduct their daily lives. The AI revolution may be compared to the Industrial Revolution, as both revolutions involve a fundamental shift in how work is done and what is produced, improving existing products and creating new ones, significantly affecting individuals' everyday lives.

The Industrial Revolution happened in two major phases. The first phase (1760-1840) was a shift from agrarian societies to industrialized, machine-driven manufacturing; the revolution was driven by steam power and coal (Library of Congress, n.d.; Kordas et al, 2022). The second phase (1840-1914) was a shift from steam-powered, coal, and wood production to steel, mass production, electricity, combustion engines, and petroleum (Majumdar, 2012; Kordas et al, 2022).

The AI revolution may have two phases:

The first phase can be defined as the digital-intelligence phase. In this phase, AI is confined within data centers, and it may evolve from narrow AI, Large Language Model (LLM), Large Reasoning Model (LRM), Agentic AI, to General Intelligence (AGI).

“AGI stands for Artificial General Intelligence, which means an AI system with general, human-level (or beyond) ability to learn, reason, and apply knowledge across a wide range of tasks and domains. AGI systems conceivably could handle novel situations, not just perform well on a single, narrow task. The term is controversial in several ways, including that different people mean different things by “human-level intelligence,” and there’s no universally accepted test, so claims are hard to verify. There are also safety and ethical concerns debated by AI experts.” (Stanford Institute for Human-Centered Artificial Intelligence, n.d.)

The second phase can be defined as the embodied and distributed intelligence phase. In this phase, AI is embodied within robots, giving it the ability to perceive, navigate, interact, and act on the physical world. It is also when AI is heavily integrated within distributed cyber-physical systems, giving it the ability to coordinate autonomous machines and software agents; make real-time decisions; and monitor and control large industrial complexes, infrastructure networks, financial processes, governmental processes, transportation systems, supply chains, and other large distributed systems. In this phase, AI may evolve from AGI to Artificial Super Intelligence (ASI).

“Artificial superintelligence (ASI) is a hypothetical software-based artificial intelligence (AI) system with an intellectual scope beyond human intelligence. At the most fundamental level, this superintelligent AI has cutting-edge cognitive functions and highly developed thinking skills more advanced than any human.” (Mucci & Stryker, 2023)

During the digital-intelligence phase that we are experiencing in its early stages today, similarly to the first phase of the Industrial Revolution, AI is beginning to reorganize production, communication, and decision-making. It may change the nature of jobs, especially white-collar jobs that rely heavily on knowledge work, by automating cognitive and knowledge tasks. This automation has the capacity to simplify complex cognitive work and allow easy access to large repositories of technical, administrative, and scientific knowledge. Human workers' roles will need to move toward oversight of AI, judgment, design, and responsibilities. For example, a junior software engineer, just a couple of years ago, was expected to do most of the coding for a software project based on directions provided by a senior software engineer. Today, a junior software engineer needs to understand how to prompt and oversee coding agents, use various coding harnesses, and verify/test/validate/comprehensively communicate the coding agents' coding outputs to a team; these tasks are significantly more difficult than just coding-based work; a couple of years ago, these tasks were mostly performed by a establish or senior software engineer, not by junior software engineer.

This implies that, to some degree, all jobs will be impacted, but more significantly, white-collar jobs. The composition of the white-collar workforce is already changing due to increasing demand for workers with high technical, managerial, analytical, and communication skills, while there is reduced demand for entry-level white-collar roles requiring repetitive administrative and basic knowledge work. Society needs to adapt as AI is being integrated into almost every aspect of everyday life, from how individuals work and interact on social media to how they communicate with chatbots, receive healthcare and educational services, access information, and make financial or personal decisions. These changes will require society to be more AI-literate, education to be transformed to accommodate AI technology, businesses to offer AI training to employees, governments to have stronger privacy protections, transparent AI decision-making, clear accountability standards, and equitable access to AI technologies.

During the embodied and distributed intelligence phase, similarly to the second phase of the Industrial Revolution, society may experience an intelligence explosion, where AI technologies may evolve into extremely powerful technologies, beyond digital environments, and into an intelligence explosion that will power physical systems and distributed cyber-physical systems such as critical infrastructure. Embodied AI could significantly change how blue-collar workers do their jobs, similar to the white-collar workers in the first phase; they may have to shift from directly performing repetitive, physically strenuous, or dangerous/hazardous tasks to supervising autonomous machines controlled by AIs, operating robotic systems, diagnosing AI/robotic problems,

performing AI/robotic maintenance, and coordinating AI/robotic teams. Consequently, the blue-collar roles may shift more toward roles requiring more technical, analytical, operational, and communication skills. The impacts of embedding AI, especially ASI, within distributed cyber-physical systems such as industrial complexes and critical infrastructure are too early to predict with confidence. However, it is a reasonable expectation that the societal effects of ASI would be extreme, as they will probably deeply embed every society's physical, economic, governmental, and technological systems.

Daniel Kokotajlo (former OpenAI super-alignment researcher) and its team have developed a predictive model where superintelligence could emerge before 2030, driven by recursive self-improvement (Kokotajlo et al, 2025). Google DeepMind CEO Demis Hassabis (2026) stated:

“The magnitude of this [AI] technology's impact will be unprecedented, perhaps 10x of the Industrial Revolution at 10x the speed. It will help us solve some of the biggest problems society faces from accelerating drug discovery to developing new clean energy sources to creating novel advanced materials.”

In other words, AI is a research accelerator that allows scientists to analyze enormous quantities of information, generate and evaluate hypotheses, design experiments, discover patterns, and explore solutions at a speed and scale that would be impossible for any human research team. Moreover, AI's recursive self-improvement could enable AI systems to improve their own design, accelerating the speed at which research is done and technology advances, resulting not only in an intelligence explosion but also in an unprecedented scientific and technological revolution.

The aspect of AI technology integration into businesses that will most significantly impact the nature of jobs and the composition of the workforce is the speed at which this transformation may occur. During previous technological revolutions, workers, educational institutions, businesses, and governments often had decades to adapt to new technologies. In contrast, this AI revolution's speed may profoundly change the nature of many jobs and the composition of the workforce. Consequently, society may have to face serious issues such as displaced workers having nowhere to go, outdated educational programs, and ineffective regulations that will affect negatively everyday life. Thus, it is essential for society not only to adopt AI technologies but also to adapt to the speed at which these technologies are developing and being integrated into workplaces. Governments, businesses, and educational institutions must anticipate these changes, continuously update workforce training and academic programs, and establish flexible regulations capable of responding to rapidly evolving AI systems.

AI is such a powerful technology that, if managed wisely, it may take us to the stars; if managed recklessly, it may ensure that we never reach them.

-Alex

References:

Hassabis, D. [@demishassabis]. (2026, July 14). *AI is already starting to deliver real-world benefits but to realise its immense promise, we have to navigate this critical* [Post]. X.
<https://x.com/demishassabis/status/2076957440109625718>

Kokotajlo, D., Alexander, S., Larsen, T., Lifland, E., & Dean, R. (2025, April 3). *AI 2027*. AI Futures Project. <https://ai-2027.com/race>

Kordas, A., Lynch, R. J., Nelson, B., & Tatlock, J. (2022). 9.1 The Second Industrial Revolution. *World history volume 2, from 1400*. OpenStax. <https://openstax.org/books/world-history-volume-2/pages/9-1-the-second-industrial-revolution>

Library of Congress. (n.d.). *The Industrial Revolution in the United States*. U.S. Government. <https://www.loc.gov/classroom-materials/industrial-revolution-in-the-united-states/>

Majumdar, S. K. (2012). Industrial revolutions. In *India's late, late industrial revolution*. Cambridge University Press. https://www.cambridge.org/core/books/abs/indias-late-late-industrial-revolution/industrial-revolutions/60FDB79FC74F5B6238F36865AB87D86F?utm_source=chatgpt.com

Mucci, T., & Stryker, C. (2023, December 14). *What is artificial superintelligence?* IBM. <https://www.ibm.com/think/topics/artificial-superintelligence>

Stanford Institute for Human-Centered Artificial Intelligence. (n.d.). *What is AGI (artificial general intelligence)?* <https://hai.stanford.edu/ai-definitions/what-is-agi-artificial-general-intelligence>